International Islamic University Chittagong

Department of Computer Science and Engineering

Mid-Term Examination, Spring-2023

Course Code: **CSE-2423** Course Title: **Database Management System** Semester: 4th

Time: 1 Hour 30 minutes

Marks: 30

			Mark	CO	DL
Q1.	**a)**	Define Database management system? What is data Abstraction? Write down the benefits of data abstraction	1+2 =3	CO1	C1
	b)	How does database system ensures atomicity and concurrency? Explain each with example	3	CO1	C2
	c)	Compare between DML and DDL. Write down the functions of different component of storage manager.	2+2 =4	CO1	C1
Q2.	**a)**	How does a query being executed through different components of DBMS? Explain.	3	CO1	C2
	b)	Consider a scenario where a student can register for courses. All student are enrolled to a specific department. Courses are taught by the teacher of the department which a student is enrolled. A teacher can be a course teacher also an advisor at the same time. Now draw a simple E-R diagram from the above scenario. Consider the necessary mapping cardinalities, participation constraints and attributes.	4	CO3	C6
		OR			
	b)	Construct an ER of the database used by the ABC super store, where persons with ID, name, age considered as either Customer or Employee. Customers may be either wholesale or retail customers. Employees could be either operational or manager. The manager could control the employees to ensure proper sell of the products to the customers. Consider the necessary mapping cardinalities, participation constraints and attributes.	4	CO3	C6
	c)	What do you understand by attributes? Write down the types of attributes. Explain at least three of them with proper example.	3	CO1	C1
Q3.	**a)**	Write down the fundamental operations of the relational algebra and explain them with proper example.	4	CO1	C2
	b)	**passenger** (pid, pname, pgender, pcity) **flight** (fid, fdate, time, src, dest) **book** (pid, fid) Consider the relational database given above, where the primary keys are underlined. Give an expression in the	3x2 =6	CO2	C3

	Marks	CO	BL

relational algebra:

1. Get the details of flights that are scheduled on both dates 01/03/2023 and 03/03/2023 at 18:00 hours.
2. Get the details of flights that are scheduled on either of the dates 05/03/2023 or 07/03/2023 and booked for passenger 'P01'
3. Find the passenger who have booked same flight as 'P02'

OR

		Marks	CO	BL
b)	**Product**(<u>P-id</u>, P-name, Unit_price) **Sales**(<u>S-id</u>,P-id,C-id, Emp_id, Data_of_Sale, Time, Quantity) **Customer** (<u>C-id</u>, C-name, Mobile, Street, City, Country) **Employee**(<u>Emp-id</u>, E-name, Designation, Commission, Salary, Mobile, City, Country)	3x2 =6	CO2	C3

Consider the relational database given above, where the primary keys are underlined. Give an expression in the relational algebra:

1) Find those Employees and their sales details who have sold more than 5000 on '20-Nov-2017'
2) Find the customers details who have served by employee 'X'
3) Find the customers who have purchased same products as 'Y' before '20-Nov-2017'.

Spring-23

1. a) **Database management system:**

DBMS is a computerized system that is liable or responsible for defining, creating, manipulating processing, accessing and using the database.

Data Abstraction:

The process or mechanism of hiding data

The benefits of data abstraction:

(i) It allows users to focus on a machine's basic functionality without having to understood its complex operation.

(ii) With data abstraction, programmers protect the inner workings of a system.

(iii) With data abstraction consumers remain unawa of the technical details and companies can reus the data without disrupting how consumers use their products.

(iv) Companies and developers may also create object having similar functions.

1.b)

Atomicity is mainly accomplished by complicated processes like journaling or through operating system calls.

Database achieve concurrency happens because another transaction inserts new records in the database table before the second query runs.

Example: imagine two number transactions T1 and T2, operating concurrently on an employee table in the database. The T1 transaction runs a query to fetch the employee records belonging the IT department.

1.c) Compare between DML & DDL:

DDL	DML
(i) It deals with how data is stored in the database and aids in definition the structure or schema of a database	(i) It enables us to manipulate the data that kept in the database, including retrieve, update & delete.
(ii) Data Definition Language is the full name of this language	(ii) Data Manipulation Language is the full name of this Language.
(iii) There is no additional categorisation for the DDL instructions.	(iii) The DML commands are divided into declarative and procedural (non-procedural DMLs).

Write the function of different component of storage manager in below:

Authorization and Integrity Manager:

It tests the integrity contraints and checks the authorization of users to access data.

2. Transaction Manager: It ensures that no kind of change will be brought to the database untill a transaction has been completed totally.

3. File manager: It manages the allocation of space on disk storage and the data structures used to represent information stored on disk.

4. Buffer Manager: It decides which data is in need to be cached in the main memory. This is very important as it defines the speed at which the database can be used.

2.a) Query executed in DBMS:

Parsing:

It is the first step. In this step a query is checked for syntax errors. Then it converts it into the parse tree.

Optimization: After doing query parsing the DBMS starts finding the most efficient way to executed the given query. The optimization process follows some factors for the query. These factors are indexing, joins and others optimization mechanism.

Evaluation :

After finding the best execution plan, the DBMS starts the execution of the optimized query. And it gives the results from the database. In this steps, DBMS can perform operations on the data. These operations are selecting the data, inserting something, updating the data and so on.

2. c)

Attributes:

In DBMS an attribute is a piece of data that describes an entity.

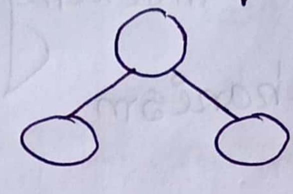
In DBMS there are various types of attributes:

~~I) Simple Attributes~~

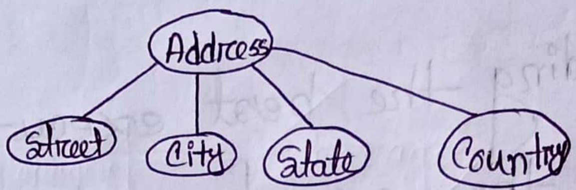
(I) Composite Attributes
(II) Multivalued "
(III) Derived Attributes
(IV) Key Attributes.

(i) Composite Attributes:

An attributes that can be split into components is a composite. It's symbol -



Ex:



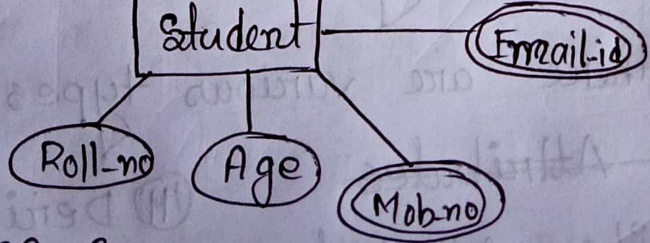
We can see in the above example, Address is a composite attributes represented by an elliptical Shape and it can be furthere subdivided into many simple attributes like street, city, state, country, Land mark etc.

(ii) Multivalued Attributes:

An attributes that can have more than one value associated with the key of the entity.

symbol:

Ex:

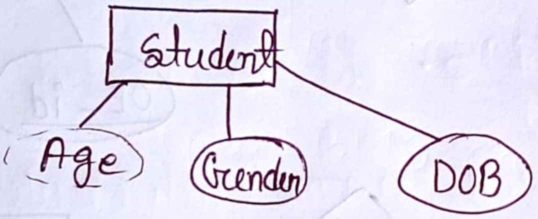


We can see in the diagram the student entity has four attributes: Roll-no, Age, Mob-no, Email-id

Here Roll-no and Age are simple as well as single valued attributes as discussed above but Mob-no & Email-id are represented by co-centric ellips are multivalued attributes.

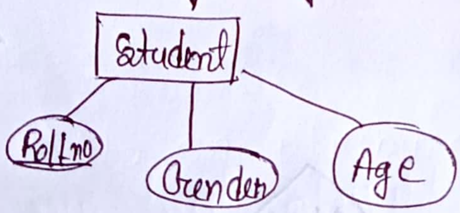
(iii) Derived Attributes: A type of attribute where the valued for that attribute will be derived from one or more of the other attribute of the same database. Symbol:

Ex:



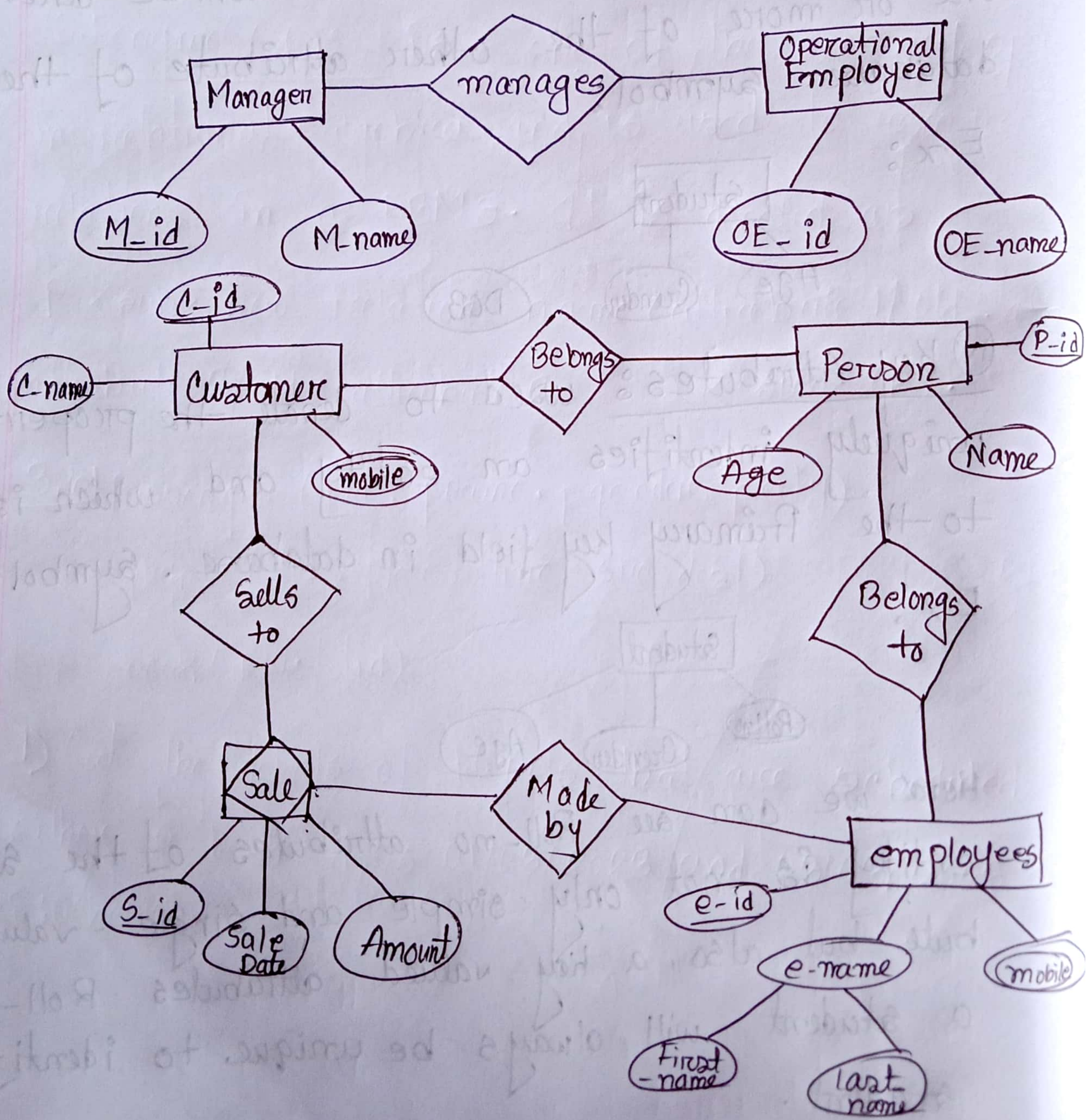
(iv) Key Attributes: used to denote the property that uniquely indentifies an entity and which is mapped to the Primary key field in database. Symbol:

Ex:



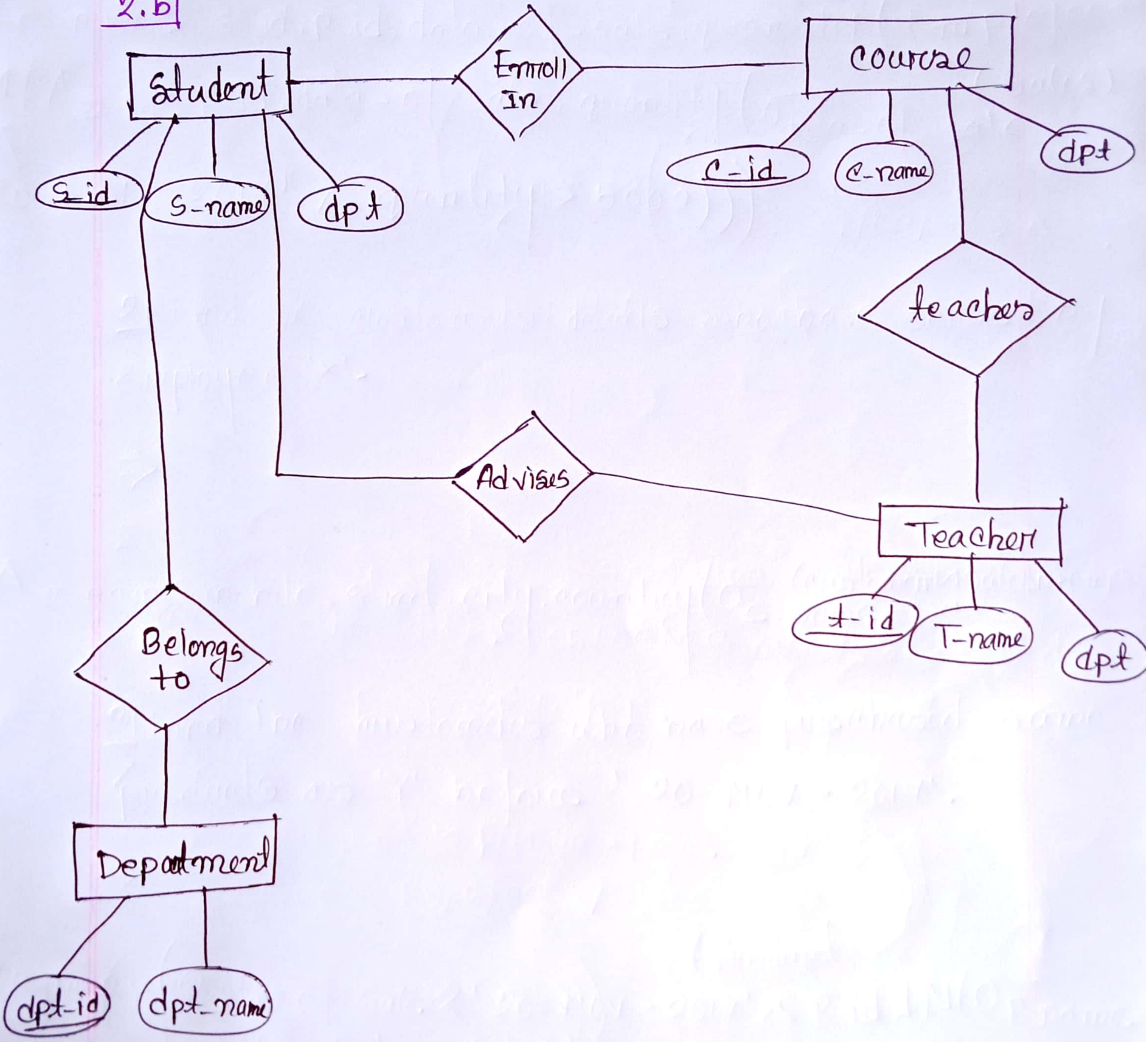
Here, we can see Roll-no attributes of the Student entity is not only simple and single-valued attribute but also, a key valued attributes. Roll-no of a student will always be unique to identify the student.

2.b] OR,



Manager
manages
Operational Employee
M_id
M_name
OE_id
OE_name
C_id
C_name
Customer
Belongs to
Person
P_id
mobile
Age
Name
Sells to
Belongs to
Sale
Made by
employees
S_id
Sale Date
Amount
e-id
e-name
mobile
First-name
last name

2.b


Student
S_id
S-name
dpt
Enroll in
Course
C-id
C-name
dpt
teaches
Advises
Teacher
t-id
T-name
dpt
Belongs to
Department
dpt-id
dpt_name

3.a) The functional operation of relational algebra:

1. Selection (σ): The operation selects rows from a relation (table) that satisfy a given condition. It is denoted as σ _{condition} (R), where R is the relation and "condition" is a logical expression.

2. Projection (π): Projection selects specific columns from a relation while discarding others. It is denoted as π _{attribute 1, attribute 2, ...} (R), where R is the relation and "attribute 1, attribute 2, -...." are the attribute (columns) to be retained.

3. Union (U): Must have same, same datatype. Duplicate row will be eliminated.

4. Intersection (∩): return common row. It is denoted as R∩S

5. Difference (-): takes the two set and returns the value that are in the first set but not the second set. It is denoted as R-S.

6. Cartesian Product (×): an operation used to merge columns from two relations. It is denoted as R×S.

7. Join (⋈): Join is a special form of cross product of two tables.

8. Rename (ρ): is one of the unary operators in relational algebra and is used to ~~rem~~ rename relations in a DBMS. It is denoted as

ρ _{new_relation_name (attribute 1/ old_attribute 2/old_attribute 2, ...)} (R).

3.b) Passenger (pid, pname, pgender, pcity)
flight (fid, fdate, time, src, dest)
book (pid, fid)

1) Get the details of flights that are scheduled on ~~either o~~ both dates 01/03/2023 and 03/03/2023 at 18:00hours.

⇒

$\pi_{fid,\ fdate,\ time,\ src,\ dest}\left(flight \bowtie \left(\sigma_{fdate='01/03/2023' \wedge time='18:00'}(flight)\right) \cap \left(\sigma_{fdate='03/03/23' \wedge time='18:00'}(flight)\right)\right)$

2) Get the details of flights that are scheduled on either of the dates 05/03/23 or 7/3/23 and booked for passenger 'P01'.

⇒

$\pi_{flight.*}\left(\sigma_{(fdate='05/3/23' \vee fdate='7/3/23')}(flight \bowtie \rho_{(pname \to pname,\ fid \to fid)})\right)$

$\cup\left(\sigma_{P\_id='P01'}(Passenger \bowtie Book)\right)$

3) Find the passenger who have booked same flight at 'P02'

⇒

$\pi_{pname}\left(\sigma_{pid='P02'}(book \bowtie \rho_{Pid \to pid}(\sigma_{pid='P01'}(book))\ flight\right)$

3.(b) Or

Product (P_id, P_name, Unit-price)

Sales (S_id, P_id, C_id, Emp_id, Data-of-sale, Time, Quantity)

customer(C_id, C_name, Mobile, street, city, country)

Employee(Emp_id, E-name, Designation, Commission, Salary, Mobile, city, country)

1) Find those Employees and their Sales ~~deo~~ details who have sold more than 5000 on '20-NOV-2017'

⇒

~~π e-name, s-id, p_id, date of sale, quantity (Employees ⋈~~

~~π s_id, p_id, data of sale, quantity (σ data_of_sale =~~

e_name, s-id, p_id, date-of-sale, quantity

(σ data_of_sale = '20-NOV-2017' $\wedge$ quantity > 5000 (Employees ⋈ (π s_id, p_id, data_of_sale,

$\pi_{e\text{-}name,\ s\text{-}id,\ p\text{-}id,\ data\text{-}of\text{-}sale,\ quantity}\Big(Employees \bowtie \big(\pi_{s\text{-}id,\ p\text{-}id,\ data\text{-}of\text{-}sale,\ quantity}(\sigma_{data\text{-}of\text{-}sale = '20\text{-}NOV\text{-}2017' \wedge quntity > 5000}(Sales))\big)\Big)$

2| Find the customers details who have served by employee 'x'.

⇒

$\pi_{c\text{-}name,\ mobile,\ street,\ city,\ country}(\sigma_{e\text{-}name = 'X'}(customer \bowtie Sales \bowtie Empl$

3| Find the customers who have purchased same products as 'Y' before '20-NOV-2017'.

⇒

$\pi_{c\text{-}name}\Big(\sigma_{data\text{-}of\text{-}sale < '20\text{-}NOV\text{-}2017' \wedge P\text{-}id\ IN\ \cancel{(Product}}\big(\sigma_{P\text{-}name = 'Y'}(Products \bowtie Sales)\big)\Big)$ $\cancel{'Y')}$